Module - 1

Basics

An electromagnetic field is a physical field produced by electrically charged particles. This field is a combination of an electric field and magnetic field.

Electric field is produced by stationary charges and magnetic field by moving charges.

Scalar qty :- Qty characterised by magnitude is called scalar qty - ex: mass, time, temp etc.

vector qty :- Both magnitude & direc'n - Force, velocity, acceleratn.

Scalar & vector field

field :- Region in which a particular physical function has a value at each & every pt in that region.

Scalar field :- If scalar qty is specified in a region - scalar field.

Atmospheric temp & pressure in a region.

Electrostatic potential.

Vector field:-

vector qty is specified in a region, then region is said to be vector field.

ex: gravitational force & wind velocity in atmosphere, Magnetic field produced by a current carrying conductor.

vector

unit vector – $\frac{\text{Vector}}{\text{Magnitude}} = \frac{\vec{OA}}{|\vec{OA}|}$

Vector mplication:-

1) **Scalar mpl'n / Dot product**

Scalar mpl'n of 2 vectors is a scalar qty whose magnitude is the product of magnitude of vectors and cosine of angle b/w them.

Dot product of $\bar{A}$ & $\bar{B}$

$$\bar{A} \cdot \bar{B} = |\bar{A}||\bar{B}| \cos\theta$$

$$\bar{A} = A_x \hat{a_x} + A_y \hat{a_y} + A_z \hat{a_z}$$

$$\bar{B} = B_x \hat{a_x} + B_y \hat{a_y} + B_z \hat{a_z}$$

$$\bar{A} \cdot \bar{B} = [A_x \hat{a_x} + A_y \hat{a_y} + A_z \hat{a_z}] \cdot [B_x \hat{a_x} + B_y \hat{a_y} + B_z \hat{a_z}]$$

$$= A_x B_x + A_y B_y + A_z B_z$$

$a_x \cdot a_x = a_y \cdot a_y = a_z \cdot a_z = 1$

$a_x \cdot a_y = a_y \cdot a_z = a_z \cdot a_x = 0.$

2) **vector product / cross product.**

It is the product of magnitude of vectors & sine of angle b/w them.

The product will be a vector & its direction is $\perp^r$ to plane containing the 2 vectors.

$$\bar{A} \times \bar{B} = |\bar{A}| \cdot |\bar{B}| \sin\theta$$

$\bar{A} = A_x \hat{a_x} + A_y \hat{a_y} + A_z \hat{a_z}$ & $B = B_x \hat{a_x} + B_y \hat{a_y} + B_z \hat{a_z}$

$$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

note: → If $\bar{A} \cdot \bar{B} = 0$ – vector are orthogonal [$\bar{A}$ & $\bar{B}$ are perpendicular]

If $\bar{A} \times \bar{B} = 0$, then vectors are solenoidal ($\bar{A}$ & $\bar{B}$ are parallel)

Q. If $\bar{A} = 2\hat{a_x} + 2\hat{a_y} - \hat{a_z}$ and $B = 6\hat{a_x} - 3\hat{a_y} + 2\hat{a_z}$

1) find $\bar{A} \cdot \bar{B}$ & 2) $\bar{A} \times \bar{B}$

1) $\bar{A} \cdot \bar{B} = 12 - 6 - 2 = 4$

2) $\bar{A} \times \bar{B} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ 2 & 2 & -1 \\ 6 & -3 & 2 \end{vmatrix} = \hat{a_x}[4-3] - \hat{a_y}[4+6] + \hat{a_z}[-6-12]$

$= \hat{a_x} - 10\hat{a_y} - 18\hat{a_z}$

? If $\bar{A} = 4\hat{a_x} + 3\hat{a_y} + \hat{a_z}$ & $\bar{B} = 2\hat{a_x} - \hat{a_y} + 2\hat{a_z}$.

find a) a unit vector $\perp^r$ to $\bar{A}$ & $\bar{B}$

b) Angle b/w $\bar{A}$ & $\bar{B}$

a) $\bar{A} \times \bar{B} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ 4 & 3 & 1 \\ 2 & -1 & 2 \end{vmatrix} = \hat{a_x}[6+1] - \hat{a_y}[8-2] + \hat{a_z}[-4-6]$

$= 7\hat{a_x} - 6\hat{a_y} - 10\hat{a_z}$

unit vector $\perp$ to $\bar{A}$ & $\bar{B}$ = $\hat{a_n} = \cancel{\frac{\bar{A}\times\bar{B}}{|\bar{A}\times\bar{B}|}}\ \frac{\bar{A}\times\bar{B}}{|\bar{A}\times\bar{B}|}$

$= \frac{7\hat{a_x} - 6\hat{a_y} - 10\hat{a_z}}{\sqrt{49+36+100}}$

$= \frac{7\hat{a_x} - 6\hat{a_y} - 10\hat{a_z}}{\sqrt{185}}$

b) $\bar{A}\cdot\bar{B} = A_xB_x + A_yB_y + A_zB_z$

$= 8 - 3 + 2 = 7$

$A\cdot B = |A||B|\cos\theta$ $\quad |A| = \sqrt{16+9+1} = \sqrt{26}$

$|B| = \sqrt{4+1+4} = \sqrt{9}$.

$\cos\theta = \frac{\bar{A}\cdot\bar{B}}{|\bar{A}|\cdot|\bar{B}|} = \frac{7}{3\sqrt{26}} = 62.76$

A.W S.T vectors $\bar{A} = 6\hat{a_x} + 2\hat{a_y} - 5\hat{a_z}$ & $\bar{B} = 5\hat{a_x} - 5\hat{a_y} + 4\hat{a_z}$ are $\perp^r$ to each other.

$A\cdot B = 0$.

$\bar{A}\cdot B = 30 - 10 - 20 = 0$.

? S.T vectors $\bar{A} = 4\hat{a_x} - 2\hat{a_y} + 2\hat{a_z}$ & $B = -6\hat{a_x} + 3\hat{a_y} - 3\hat{a_z}$ are $\parallel^{el}$ to each other

$\bar{A}\times\bar{B} = 0$ $\qquad \bar{A}\times\bar{B} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ 4 & -2 & 2 \\ -6 & 3 & -3 \end{vmatrix}$

$= \hat{a_x}[+6-6] - \hat{a_y}[-12+12] + \hat{a_z}[12-12] = 0$

vectors are $\parallel^{el}$.

Coordinate system :-

It is a s/m which is used to represent a pt in space.

3 type of coordinate s/ms are

1) Rectangular / cartesian s/m.

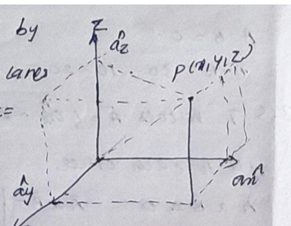
2) cylindrical coordinate s/m

3) spherical coordinate s/m.

Rectangular / cartesian co-ordinate s/m.

In this s/m 3 co-ordinate axes x, y, z are mutually $\perp^r$ to each other; 3 axes intersect at a common pt called origin of the s/m.

A point P is defined by the intersection of 3 planes $x = \text{const}$, $y = \text{const}$ & $z = \text{const}$ planes.



Variation of x, y, z axis

$-\infty \le x \le \infty$

$-\infty \le y \le \infty$

$-\infty \le z \le \infty$.

Position vector of point P

~~$\bar{r} = \bar{op} = x \hat{a}_x$~~ $\bar{r}_{op} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$

magnitude $|\bar{r}_{op}| = \sqrt{(x_1)^2 + (y_1)^2 + (z_1)^2}$

Differential elements

consider a pt $P(x, y, z)$ in rectangular co-ordinate sym

lets increase each coordinate by a differential amt. A point P' is obtained with coordinates $(x+dx, y+dy, z+dz)$

dx - differential length in x direcn

dy - differential length in y direcn

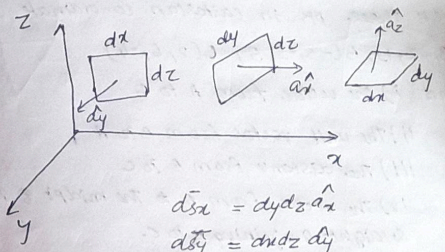
dz - differential length in z direcn.

Differential length

$$d\bar{l} = dx\,\hat{a}_x + dy\,\hat{a}_y + dz\,\hat{a}_z$$

$$|d\bar{l}| = \sqrt{dx^2 + dy^2 + dz^2}$$

Differential surface area :-



$$d\bar{s}_x = dy\,dz\,\hat{a}_x$$

$$d\bar{s}_y = dx\,dz\,\hat{a}_y$$

$$d\bar{s}_z = dx\,dy\,\hat{a}_z$$

Diff. vol :-

$dv = dx\,dy\,dz$

Diff. volume is a rectangular parallelopiped.

1.2 ? Two pts $A(2,2,1)$ & $B(3,-4,2)$ are given in the cartesian system. Obtain the vector from A to B & a unit vector directed from A to B.

$\bar{A} = 2\bar{a}_x + 2\bar{a}_y + \bar{a}_z$ & $\bar{B} = 3\bar{a}_x - 4\bar{a}_y + 2\bar{a}_z$

$\bar{AB} = \bar{B} - \bar{A} = (3-2)a_x + (-4-2)\bar{a}_y + (2-1)\bar{a}_z$

$\bar{AB} = \bar{a}_x - 6\bar{a}_y + \bar{a}_z$

unit vector

$$\bar{a}_{AB} = \frac{\overline{AB}}{|\overline{AB}|} = \frac{\bar{a}_x - 6\bar{a}_y + \bar{a}_z}{6.164} = 0.16\bar{a}_x - 0.97\bar{a}_y + 0.162\bar{a}_z$$

$|\bar{a}_{AB}| = \underline{\underline{1}}$

1.3 Given three pts in cartesian co-ordinate s/m as $A(3,-2,1)$, $B(-3,-3,5)$, $C(2,6,-4)$.

Find i) The vector from A to C

ii) The unit vector from B to A.

iii) The distance from B to C

iv) The vector from A to the midpt of the straight line joining B to C.

Position vectors are

$\bar{A} = 3\hat{a_x} - 2\hat{a_y} + \hat{a_z}$, $\bar{B} = -3\hat{a_x} - 3\hat{a_y} + 5\hat{a_z}$,

$C = 2\hat{a_x} + 6\hat{a_y} - 4\hat{a_z}$

i) Vector from A to C

$\overline{AC} = \bar{C} - \bar{A} = -\hat{a_x} + 8\hat{a_y} - 5\hat{a_z}$

ii) $\overline{BA} = \bar{A} - \bar{B} = 6\hat{a_x} + \hat{a_y} - 4\hat{a_z}$

$|\overline{BA}| = \sqrt{6^2 + 1^2 + 16} = 7.28$

$\bar{a}_{BA} = \frac{\overline{BA}}{|\overline{BA}|} = 0.82\hat{a_x} + 0.13\hat{a_y} - 0.54\hat{a_z}$

iii) Distance b/w B & C

$\overline{BC} = \bar{C} - \bar{B} = 5\hat{a_x} + 9\hat{a_y} - 9\hat{a_z}$

~~$\overline{BC} = 5\hat{a_x} + 9\hat{a_y}$~~ $-9\hat{a_z}$

Distance $= \sqrt{5^2 + 9^2 + (-9)^2} = 13.67$.

iv) $B(x_1, y_1, z_1)$ & $C(x_2, y_2, z_2)$

midpoint of BC $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$.

Midpoint of BC $= \left(\frac{-3+2}{2}, \frac{-3+6}{2}, \frac{5-4}{2}\right)$.

$= (-0.5, 1.5, 0.5)$

vector from A to this mid pt

$= [-0.5-3]\bar{a}_x + [1.5+2]\hat{a_y} + [+0.5-1]\hat{a_z}$

$= -3.5\hat{a_x} + 3.5\hat{a_y} - 0.5\hat{a_z}$.

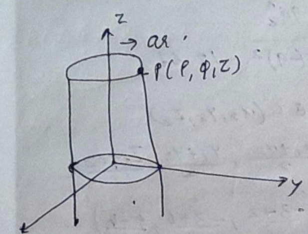
cylindrical co-ordinate s/m.

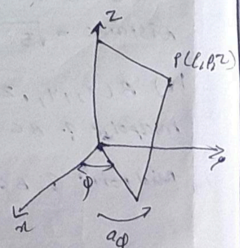
Consider a pt P in the cylindrical coordinate s/m which is the intersection of 3 mutually perpendicular surfaces. They are:

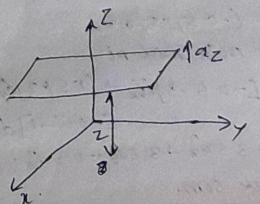
i) cylinder of radius 'ρ' with z axis as the axis of cylinder

ii) a half plane perpendicular to the x-y plane and at an angle 'ϕ' with x-~~axis~~ respective plane.

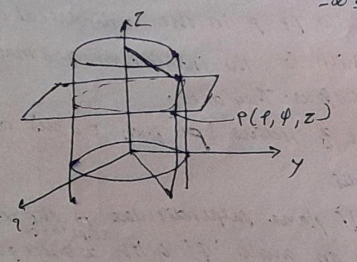
iii) another plane with const 'z' which is ||el to x-y plane.

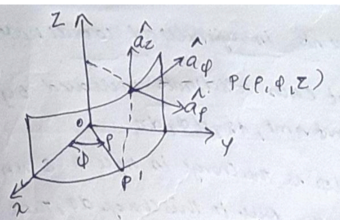






The ranges of the variables are
$0 \le r \le \infty$
$0 \le \phi \le 2\pi$
$-\infty \le z \le \infty$





Similar to cartesian coordinate system there are 3 unit vectors in the ρ, φ, & z directions denoted as $\hat{a}_\rho$, $\hat{a}_\phi$ and $\hat{a}_z$

$\bar{a}_\rho$ is $\perp^r$ to surface of cylinder at a given pt coming radially outward. lies in a plane ‖el to xy plane ~~bt it is tangent to the cylinder at a given pt coming radially outward~~

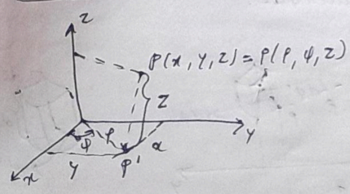
$\bar{a}_\phi$ lies parallel to xy plane bt it is tangent to cylinder & pointing in a direction of incg ϕ at the given pt.

$\bar{a}_z$ is ‖el to z axis and directed twds increasing z

Vector P can be represented as

$$\bar{P} = P_\rho \bar{a}_\rho + P_\phi \bar{a}_\phi + P_z \bar{a}_z$$

<u>Relations b/w cylindrical & cartesian s/ms.</u>



$\rho = \sqrt{x^2 + y^2}$
$x = \rho \cos\phi$
$y = \rho \sin\phi$
$z = z$
$\phi = \tan^{-1}\frac{y}{x}$

Differential elements in cylindrical coordinate s/m:

$P(r, \phi, z)$ - Let each coordinate increased by differential ~~element~~ amt, dr, $d\phi$, dz

In ϕ direcn, $d\phi$ is the change in angle ϕ & isn't the differential length. Due to this change $d\phi$, there exist a differential arc length in ϕ direcn. This arc length is $r d\phi$.

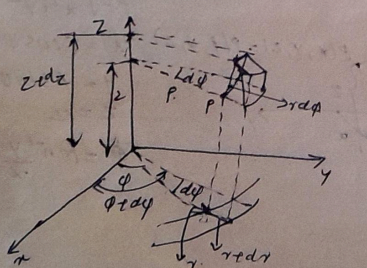
dr - Differential length in r direcn
$r d\phi$ - " " ϕ "
dz - " " z "

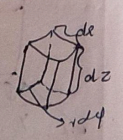
$$dL = dr\,\bar{a}_r + r\,d\phi\,\bar{a}_\phi + dz\,\bar{a}_z$$

The magnitude of diff. length $= \sqrt{(dr)^2 + (r\,d\phi)^2 + (dz)^2}$

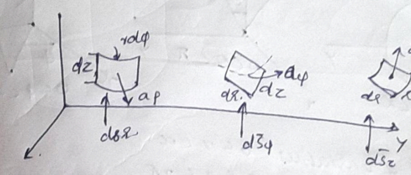
Differential volume

$$dV = r\,dr\,d\phi\,dz.$$





Differential Surface areas



$d\bar{s}_r$ = Diff. vector surface area normal to r direcn
$= r\,d\phi\,dz\,\bar{a}_r$

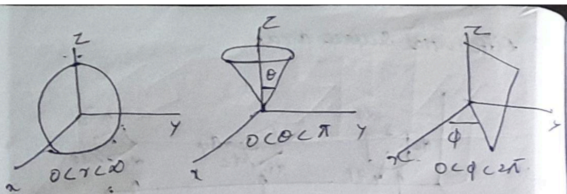
$d\bar{s}_\phi$ = Diff. vector surface area normal to ϕ direcn
$= dr\,dz\,\bar{a}_\phi$

$d\bar{s}_z$ = Diff. vector surface area normal to z direcn
$= r\,dr\,d\phi\,\bar{a}_z$

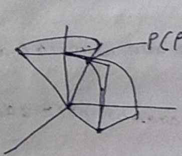
Spherical co-ordinate s/m:

Surfaces used to define the spherical coordinate s/m on 3 cartesian axes are

1) sphere of radius r, origin as centre of sphere
2) Right circular cone with apex at origin & its axis as z axis. Its half angle is θ. It rotates about z axis and θ varies from 0 to π.
3) A half plane $\perp^r$ to xy plane containing z axis making an angle ϕ with xz plane.



Ranges of variables $0 \le r < \infty$
$0 \le \phi \le 2\pi$
$0 \le \theta \le \pi$



Vector
$\bar{P} = P_\rho \hat{a}_\rho + P_\theta \hat{a}_\theta + P_\phi \hat{a}_\phi$

$\hat{a}_\rho$ → unit vector directed from centre of sphere radially outwards

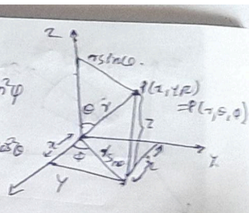
$\hat{a}_\theta$ – unit vector tangentially to the sphere & oriented in +ve direction of θ

$\hat{a}_\phi$ – unit vector tangentially to both the sphere and conical surface & oriented in the direction of +ve ϕ

Relation b/w cartesian & spherical s/ms

$x = r\sin\theta\cos\phi \quad y = r\sin\theta\sin\phi \quad z = r\cos\theta$

$r^2 = x^2 + y^2 + z^2$
$= r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi + r^2\cos^2\theta$
$= r^2\sin^2\theta[\sin^2\phi + \cos^2\phi) + r^2\cos^2\theta$
$= r^2(1)$



$r = \sqrt{x^2+y^2+z^2}$
$\tan\phi = y/x \quad \cos\theta = z/r$

While using the formulae care must be taken to place the angles θ & ϕ in the correct quadrants acc to signs of x, y, z

Differential elements in spherical co-ordinate s/m

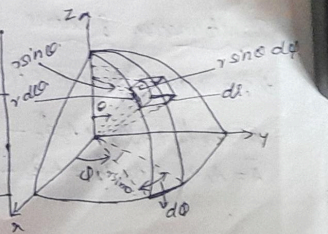
$P(\rho, \theta, \phi)$

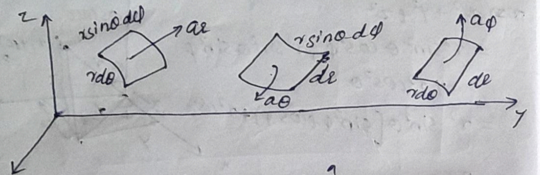
Differential length:-

$dl = d\rho\,\hat{a}_\rho + \rho\,d\theta\,\hat{a}_\theta + \rho\sin\theta\,d\phi\,\hat{a}_\phi$

* diff. vol.
$dv = d\rho \times \rho\,d\theta \times \rho\sin\theta\,d\phi$
$= \rho^2\sin\theta\,d\rho\,d\theta\,d\phi$

Differential increments in coordinates	corresponding change in lengths
dr	dr
$d\theta$	$r\,d\theta$
$d\phi$	$r\sin\theta\,d\phi$





$ds_r = r^2 \sin\theta \, d\theta \, d\phi \, \hat{a_r}$

$ds_\theta = r \sin\theta \, dr \, d\phi \, \hat{a_\theta}$

$ds_\phi = r \, dr \, d\theta \, a_\phi$

? Given pt $P(-2, 6, 3)$. Express in spherical coordinate sim.

$x = -2,\ y = 6,\ z = 3$

$r = \sqrt{4 + 36 + 9} = 7$

$\theta = \cos^{-1} \frac{z}{r} = \frac{3}{7} = 64.2$

$\phi = \tan^{-1} y/x = \tan^{-1} -3 =$

Transformation of vectors

1) Transformation of vectors from cartesian to cylindrical

$\bar{A} = A_x \hat{a_x} + A_y \hat{a_y} + A_z \hat{a_z}$ — $\bar{A}$ in cartesian coordinates.

$\bar{A} = A_r \hat{a_r} + A_\phi \hat{a_\phi} + A_z \hat{a_z}$ — $\bar{A}$ in cylindrical ..

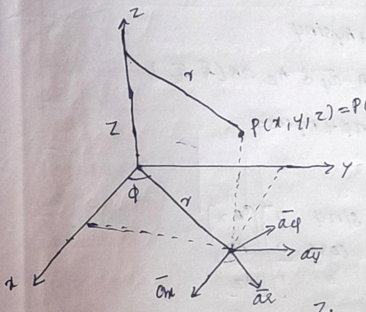
component of vector in the dirn of any unit vector is dot product with that unit vector. Hence component of $\bar{A}$ in the dirn of $\bar{a_r}$ $\bar{A} \cdot \bar{a_r}$.

$A_r = (\bar{A} \cdot \bar{a_r})$

$= [A_x \bar{a_x} + A_y \bar{a_y} + A_z \bar{a_z}] \cdot \bar{a_r}$

$= A_x \bar{a_x} \cdot \bar{a_r} + A_y \bar{a_y} \cdot \bar{a_r} + A_z \bar{a_z} \cdot \bar{a_r}$

magnitude of all unit vectors is unity, hence it is necessary to find angle b/w unit vectors to obtain various dot products.



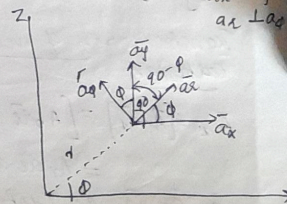
$a_x \perp a_y$
$a_r \perp a_\phi$

$\angle$ b/w $\bar{a_x}$ & $\bar{a_r}$ is ϕ

$\angle$ b/w $\bar{a_y}$ & $\bar{a_r}$ is $90 - \phi$

$\angle$ b/w $\bar{a_x}$ & $\bar{a_\phi}$ is $90 + \phi$

$\angle$ b/w $\bar{a_y}$ & $\bar{a_\phi}$ is ϕ



$\bar{a_x} \cdot \bar{a_r} = \cos\phi$

$\frac{S \mid A}{T \mid C}$

$a_x \cdot a_\phi = \cos(90+\phi)$

$= -\sin\phi$

$\bar{a_y} \cdot \bar{a_r} = \cos(90-\phi)$

$= \sin\phi$

$\sin 45$
$\cos 45$
$\tan 45$

$\bar{a_y} \cdot \bar{a_\phi} = \cos\phi$

$\bar{a_z} \cdot \bar{a_r} = \bar{a_z} \cdot \bar{a_\phi} = 0$ as $\bar{a_z}$ is perpendicular to $\bar{a_r}$ & $\bar{a_\phi}$

& $\bar{a_z} \cdot \bar{a_z} = 1$

$A_r = A_x \cos\phi + A_y \sin\phi$

Smlrly A_ϕ $(\bar{A} \cdot \bar{a_\phi}$ & A_z as $(\bar{A} \cdot \bar{a_z})$

$A_\phi = -A_x \sin\phi + A_y \cos\phi$

$A_z = A_z$

$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Transformation of vector from cylindrical to cartesian

$\bar{A}_x = [\bar{A} \cdot \bar{a_x}] = [A_r \bar{a_r} + A_\phi \bar{a_\phi} + A_z \bar{a_z}] \cdot \bar{a_x}$

$= A_r \bar{a_r} \cdot \bar{a_x} + A_\phi \bar{a_\phi} \cdot \bar{a_x} + A_z \bar{a_z} \cdot \bar{a_x}$

$\bar{a_r} \cdot \bar{a_x} = \bar{a_x} \cdot \bar{a_r}$ (dot product is commutative)

$= \cos\phi$

$A_x = A_r \cos\phi - A_\phi \sin\phi$

$A_y = A_r \sin\phi + A_\phi \cos\phi$

$A_z = A_z$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}$$

Dot operator	$\bar{a_r}$	$\bar{a_\phi}$	$\bar{a_z}$
$\bar{a_x}$	$\cos\phi$	$-\sin\phi$	0
$\bar{a_y}$	$\sin\phi$	$\cos\phi$	0
$\bar{a_z}$	0	0	1

? Transform the vector field $\bar{w} = 10\bar{a_x} - 8\bar{a_y} + 6\bar{a_z}$ to cylindrical cordinate sLm at pt $P(10,8,6)$

$w_x = 10 \quad w_y = -8 \quad w_z = 6$

$\cancel{w_x =}$ $$\begin{bmatrix} w_r \\ w_\phi \\ w_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}$$

$\phi = \tan^{-1} y/x$

$= \tan^{-1}\left[\frac{-8}{10}\right]$

$= -38.65$

$\phi \rightarrow 4^{th}$ quadrant

$\cancel{\phi = 360 - 321}$

$360 - 38.65$

$= 321.35$

$\phi = 360 - 38.65 = 321.34.$

$P = \sqrt{x^2 + y^2} = \sqrt{10^2 + 8^2} = 12.8$

$w_P =$

$$w_\rho = w_x \cos\phi + w_y \sin\phi + 0$$

$$= 10\cos 321.34 - 8\sin 321.34$$

$$= 12.80$$

$$w_\phi = -w_x \sin\phi + w_y \cos\phi + 0$$

$$= -10\sin 321.34 - 8\cos 321.34 =$$

$$= 0$$

$$w_z = 6$$

$$\bar{w} = 12.801\, \hat{a}_\rho + 6\,\hat{a}_z$$

? Given pt $(-2,6,3)$ and vector $\bar{A} = y\hat{a}_x + (x+z)\hat{a}_y$.
Evaluate $\bar{A}$ at pt P in cylindrical coordinate s/m

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\Phi = \tan^{-1} y/x = -71.56$$

$$\phi = 180 + (-71.56) = 108.4$$

$$\bar{A} = y\hat{a}_x + [x+z]\hat{a}_y + 0$$

$A_x = y \qquad A_y = x+z \qquad A_z = 0.$

$$A_\rho = A_x \cos\phi + A_y \sin\phi$$

$$= y\cos 108.4 + (x+z)\sin(108.4)$$

$$= -0.945$$

$$A_\phi = -A_x \sin\phi + A_y \cos\phi$$

$$= -y\sin\phi + (x+z)\cos\phi$$

$$= -5.7$$

$$A_z = 0$$

$$A_\rho = -0.94\,\hat{a}_\rho - 5.7\,\hat{a}_\phi$$

H.W Given the cartesian coordinates of vector field
$\bar{H} = 20\hat{a}_\rho - 10\hat{a}_\phi + 3\hat{a}_z$ at pt $(x=5, y=2, z=-1)$

$$\begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} H_\rho \\ H_\phi \\ H_z \end{bmatrix}$$

$$\phi = \tan^{-1} y/x = \tan^{-1}(2/5) = 21.8$$

$$H_x = H_\rho \cos\phi - H_\phi \sin\phi$$

$$= 20\cos 21.8 + 10\sin 21.8 = \cancel{48.06}$$

$$= 22.282$$

$$H_y = H_\rho \sin\phi + H_\phi \cos\phi = -1.856.$$

$$H_z = H_z = 3$$

Transformation of vectors from cartesian to spherical :-

$$\bar{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z.$$

To transform it into spherical s/m. component

$\bar{A}$ in $\bar{a_r}$ direc-n

$$A_r = \bar{A}\cdot\bar{a_r} = [A_x\hat{a_x} + A_y\hat{a_y} + A_z\hat{a_z}]\cdot\hat{a_r}$$

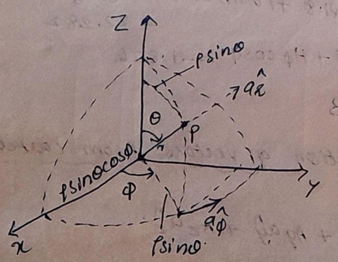
$$= A_x(\hat{a_x}\cdot\hat{a_r}) + A_y(a_y\cdot\hat{a_r}) + A_z(\hat{a_z}\cdot\hat{a_r})$$

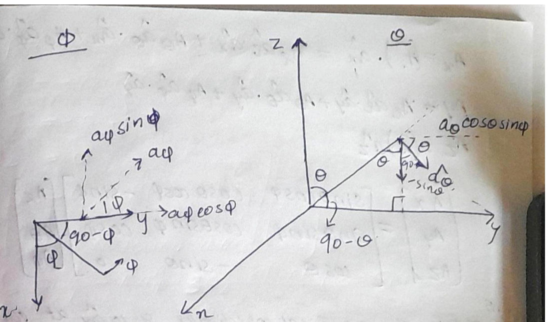
$$A_\theta = \bar{A}\cdot\bar{a_\theta} = [A_x\hat{a_x} + A_y\hat{a_y} + A_z\hat{a_z}]\cdot\hat{a_\theta}$$

$$= A_x\hat{a_x}\cdot\hat{a_\theta} + A_y\hat{a_y}\cdot\hat{a_\theta} + A_z\hat{a_z}\cdot\hat{a_\theta}$$

$$A_\phi = \bar{A}\cdot\hat{a_\phi} = A_x\hat{a_x}\cdot\hat{a_\phi} + A_y\hat{a_y}\cdot\hat{a_\phi} + A_z\hat{a_z}\cdot\hat{a_\phi}$$

	$\hat{a_r}$	$\hat{a_\theta}$	$\hat{a_\phi}$
$\hat{a_x}$	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$
$\hat{a_y}$	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$
$\hat{a_z}$	$\cos\theta$	$-\sin\theta$	0





dot operator	$\hat{a_r}$	$\hat{a_\theta}$	$\hat{a_\phi}$
$\hat{a_x}$	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$
$\hat{a_y}$	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$
$\hat{a_z}$	$\cos\theta$	$-\sin\theta$	0

Cartesian to spherical

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Trformn of vectors from spherical to cartesian.

$$\bar{A} = A_r\hat{a_r} + A_\theta\hat{a_\theta} + A_\phi\hat{a_\phi}$$

$$A_x = (\bar{A}) \cdot \hat{a}_x = A_r \hat{a}_r \cdot \hat{a}_x + A_\theta \hat{a}_\theta \cdot \hat{a}_x + A_\phi \hat{a}_\phi \cdot \hat{a}_x$$

$$A_y = A_r \hat{a}_r \cdot \hat{a}_y + A_\theta \hat{a}_\theta \cdot \hat{a}_y + A_\phi \hat{a}_\phi \cdot \hat{a}_y$$

$$A_z = (\bar{A}) \hat{a}_z$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Ex 1.15 Obtain the spherical coordinates of $10\hat{a}_x$ at the point $P(-3, 2, 4)$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Given vector $\bar{A} = 10\hat{a}_x$.

$$F_r = \bar{F} \cdot \hat{a}_r = 10\hat{a}_x \cdot \hat{a}_r$$
$$= 10\sin\theta\cos\phi$$

$$\phi = \tan^{-1}\frac{y}{x} = \tan^{-1}\left(\frac{2}{-3}\right) = -33.42$$

$\phi = 180 - 33.42 = 146.57$; $\cos\phi = -0.837$

$\sin\phi = 0.554$

$$\theta = \cos^{-1} z/r =$$

$$r = \sqrt{x^2+y^2+z^2} = \sqrt{9+4+16} = 5.385$$

$$\theta = \frac{4}{5.38} = 42.03$$

$\cos\theta = 0.742$ $\sin\theta = 0.669$

$$F_r = \cancel{10\sin} \; 10 \times 0.669 \times \cancel{0.554} \; -(0.832)$$
$$= -5.57$$

$$F_\theta = 10\hat{a}_x \cdot \hat{a}_\theta = 10\cos\theta\cos\phi$$
$$= 10 \times 0.742 \times -(0.832)$$
$$= -6.18$$

$$F_\phi = -10 \times 0.554 = -5.547$$

$$\bar{F} = -5.57\hat{a}_r - 6.18\hat{a}_\theta - 5.54\hat{a}_\phi$$

1.16 Express $\bar{B} = r^2\hat{a}_r + \sin\theta\,\hat{a}_\phi$ in the cartesian co-ordinates. Obtain $\bar{B}$ at $P(1, 2, 3)$

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix}$$

$$B_x = B_r \sin\theta\cos\phi - \sin\theta\sin\phi$$

$$r = \sqrt{x^2+y^2+z^2} = \sqrt{14} = 3.74$$

$$\cos\theta = z/r = \frac{3}{3.74}$$

$$\theta = 36.7$$

$$\tan\phi = y/x = 2$$

$$\phi = 63.43$$

$$B_x = (3.74)^2 \times \sin 36.7 \cos 63.43 - \sin 36.7 \times \sin 63.43$$

$= 8.204$

$B_y = r^2 \sin\theta \sin\phi + \cos\phi \sin\theta$

$= 7.748$

$B_z = r^2 \cos\theta = 11.21$

Transformation of vectors from spherical to cylindrical

$\bar{A} = A_r \hat{a_r} + A_\theta \hat{a_\theta} + A_\phi \hat{a_\phi}$

$\bar{A} = A_\rho \hat{a_\rho} + A_\phi \hat{a_\phi} + A_z \hat{a_z}$

$A_\rho = A_r \hat{a_r} \cdot \hat{a_\rho} + A_\theta \hat{a_\theta} \cdot \hat{a_\rho} + A_\phi \hat{a_\phi} \cdot \hat{a_\rho}$

$A_\phi = A_r \hat{a_r} \cdot \hat{a_\phi} + A_\theta \hat{a_\theta} \cdot \hat{a_\phi} + A_\phi \hat{a_\phi} \cdot \hat{a_\phi}$

$A_z = A_r \hat{a_r} \cdot \hat{a_z} + A_\theta \hat{a_\theta} \cdot \hat{a_z} + A_\phi \hat{a_\phi} \cdot \hat{a_z}$

	$\hat{a_r}$	$\hat{a_\theta}$	$\hat{a_\phi}$
$\hat{a_\rho}$	$\sin\theta$	$\cos\theta$	0
$\hat{a_\phi}$	0	0	1
$\hat{a_z}$	$\cos\theta$	$-\sin\theta$	0

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Transformation of vectors from cylindrical to spherical

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

Del operator :-

'∇' is called vector differential operator

cartesian coordinate s/m

$$\nabla = \frac{\partial}{\partial x} \hat{a_x} + \frac{\partial}{\partial y} \hat{a_y} + \frac{\partial}{\partial z} \hat{a_z}$$

cylindrical coordinate s/m

$$\nabla = \frac{\partial}{\partial \rho} \hat{a_\rho} + \frac{1}{\rho} \frac{\partial}{\partial \phi} \hat{a_\phi} + \frac{\partial}{\partial z} \hat{a_z}$$

spherical coordinate s/m

$$\nabla = \frac{\partial}{\partial r} \hat{a_r} + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{a_\theta} + \frac{1}{r\sin\theta} \frac{\partial}{\partial \phi} \hat{a_\phi}$$

Gradient of a scalar field [∇V]

1. Gradient of a scalar $\bar{F}$, $\nabla F \rightarrow$ vector
2. Divergence of vector $\bar{F}$, $\nabla \cdot \bar{F} \rightarrow$ Scalar
3. Curl of vector $\bar{F} \rightarrow \nabla \times \bar{F} \rightarrow$ Vector

Gradient of a scalar field V is a vector that represents both the magnitude & the direction of maximum space rate of inc of V.

For cartesian coordinate sim

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$$

For cylindrical coordinate sim

$$\nabla V = \frac{\partial V}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{a}_\phi + \frac{\partial V}{\partial z}\hat{a}_z$$

For spherical coordinate

$$\nabla V = \frac{\partial V}{\partial r} + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a}_\phi$$

<u>Properties of gradient.</u>

1) Magnitude of ∇V = maximum rate of change in V per unit distance.

2) ∇V points in the direction of maximum rate of change in V.

3) If $\boxed{\bar{A} = \nabla V}$, then V is said to be scalar potential of $\bar{A}$

4) The component of ∇V in the direction of a unit vector $\hat{a}_n$ is $\boxed{\nabla V \cdot \hat{a}_n}$ and is called directional derivative of V along $\bar{a}$

? Find the gradient of following scalar fields

a) $V = e^{-z}\sin 2x \cosh y$

b) $U = \rho^2 z \cos 2\phi$

c) $W = 10r^4 \sin\theta \cos\phi$

d) $F = 6\rho^4 z^3 \sin\phi$.

a) $$\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$$

hyperbolic fun

$$= e^{-z}\cosh y \, 2\cos 2x \, \hat{a}_x + e^{-z}\sin 2x \sinh y \, \hat{a}_y - e^{-z}\sin 2x \sinh y \, \hat{a}_z$$

$$= 2e^{-z}\cosh y \cos 2x \, \hat{a}_x + e^{-z}\sin 2x \sinh y \, \hat{a}_y - e^{-z}\sin 2x \sinh y \, \hat{a}_z$$

b) $$\nabla U = \frac{\partial V}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{a}_\phi + \frac{\partial V}{\partial z}\hat{a}_z$$

$$= 2\rho z \cos 2\phi \, \hat{a}_\rho + \frac{1}{\rho}\rho^2 z \, {-\sin 2\phi} \cdot 2 \, \hat{a}_\phi + \rho^2 \cos 2\phi \, \hat{a}_z$$

$$= 2\rho z \cos 2\phi \, \hat{a}_\rho - 2\rho z \sin 2\phi \, \hat{a}_\phi + \rho^2 \cos 2\phi \, \hat{a}_z$$

c) $$\nabla W = \frac{\partial W}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial W}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial W}{\partial \phi}\hat{a}_\phi$$

$$= 40r^3 \sin\theta\cos\phi \, \hat{a}_r + \frac{1}{r}10r^{\cancel{4}\,3}\cos\theta\cos\phi \, \hat{a}_\theta + \frac{1}{r\cancel{\sin\theta}}10r^{\cancel{4}\,3}\cancel{\sin\theta} \, {-\sin\phi} \, \hat{a}_\phi$$

$$= 40r^3 \sin\theta\cos\phi \, \hat{a}_r + 10r^3 \cos\theta\cos\phi \, \hat{a}_\theta - 10r^3 \sin\phi \, \hat{a}_\phi$$

d) $\nabla f = \frac{\partial f}{\partial \rho} \hat{a_\rho} + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \hat{a_\phi} + \frac{\partial f}{\partial z} \hat{a_z}$

$$= 20\rho^3 z^3 \sin\phi \, \hat{a_\rho} + \frac{1}{\rho} 5\rho^4 z^3 \cos\phi \, \hat{a_\phi} + 5\rho^4 3 z^2 \sin\phi \, \hat{a_z}$$

$$= 20\rho^3 z^3 \sin\phi \, \hat{a_\rho} + 5\rho^3 z^3 \cos\phi \, \hat{a_\phi} + 15\rho^4 z^2 \sin\phi \, \hat{a_z}$$

? Find the gradient of scalar function $V = \rho^2 \sin 2\phi$ in cylindrical coordinate sim & directional derivative in the direction of vector $\bar{A} = \hat{a_\rho} + \hat{a_\phi}$ at the point $[2, \pi/4, 0]$

$$V = \rho^2 \sin 2\phi$$

$$\nabla V = \frac{\partial V}{\partial \rho} \hat{a_\rho} + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{a_\phi} + \frac{\partial V}{\partial z} \hat{a_z}$$

$$= 2\rho \sin 2\phi \, \hat{a_\rho} + \frac{1}{\rho} \cdot \rho^2 \cos 2\phi \cdot 2 + 0$$

At $P[2, \pi/4, 0]$

$$\nabla V = 4 \sin 2 \times \frac{\pi}{4} \hat{a_\rho} + 2 \times \cos 2 \times \frac{\pi}{4} \cdot 2$$

$$= 4 \hat{a_\rho}$$

Directional derivative

$$= \nabla V \cdot \hat{a_n}$$

$$= 4 \hat{a_\rho} \cdot \left[\frac{\hat{a_\rho} + \hat{a_\phi}}{\sqrt{2}} \right] = \frac{4}{\sqrt{2}} = \underline{2\sqrt{2}}$$

? Given $\phi = xy + yz + xz$. Find the gradient of ϕ at pt $(1,2,3)$ and the directional derivative of ϕ at the same point in the direction of point $(3,4,4)$.

$$\nabla \phi = \frac{\partial \phi}{\partial x} \hat{a_x} + \frac{\partial \phi}{\partial y} \hat{a_y} + \frac{\partial \phi}{\partial z} \hat{a_z}$$

$$= (y+z) \hat{a_x} + (x+z) \hat{a_y} + (y+x) \hat{a_z}$$

$\nabla \phi$ at $(1,2,3)$

$$= 5 \hat{a_x} + 4 \hat{a_y} + 3 \hat{a_z}$$

vector joining pts $(1,2,3)$ and $(3,4,4)$

$$\bar{A} = 2\hat{a_x} + 2\hat{a_y} + \hat{a_z}$$

Directional derivative

$$= \nabla \phi \cdot \hat{a_n}$$

$$= [5\hat{a_x} + 4\hat{a_y} + 3\hat{a_z}] \cdot \frac{[2\hat{a_x} + 2\hat{a_y} + \hat{a_z}]}{\sqrt{4+4+1}}$$

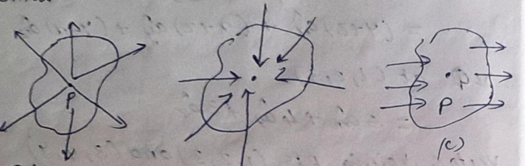
$$= \frac{10 + 8 + 3}{3} = \underline{\underline{7}}$$

<u>Divergence:-</u>

Divergence of a vector quantity $\bar{A}$ is the outward flow of vector qty per unit volume from a closed surface surrounding the volume as the volume tends to zero.

$$\boxed{\operatorname{div}\bar{A} = \nabla\cdot\bar{A} = \lim_{\Delta v \to 0} \frac{\oint \bar{A}\cdot d\bar{s}}{\Delta v}}$$

Divergence of a vector at any point may be +ve if the field lines are diverging/coming out from a small volume surrounding the point. (Fig a)



(a) (b) (c)

If the field lines are converging into a small volume the divergence of vector is −ve (Fig (b))

If the rate at which field lines are entering into a small volume surrounding the pt = to the rate at which these are leaving the same volume then the divergence of the vector is zero (Fig c)

$\bar{A} = A_x\hat{a_x} + A_y\hat{a_y} + A_z\hat{a_z}$

Del operator $\nabla = \frac{\partial}{\partial x}\hat{a_x} + \frac{\partial}{\partial y}\hat{a_y} + \frac{\partial}{\partial z}\hat{a_z}$

Divergence $\nabla\cdot\bar{A} = \frac{\partial}{\partial x}A_x + \frac{\partial}{\partial y}A_y + \frac{\partial}{\partial z}A_z$

Divergence of a vector qty is a scalar.

Cylindrical coordinate s/m.

$$\nabla\cdot\bar{A} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial}{\partial\phi}A_\phi + \frac{\partial}{\partial z}A_z$$

Spherical coordinate s/m.

$$\nabla\cdot\bar{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(A_\theta\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

Properties of divergence of vector field.

1) Divergence produces a scalar field
Divergence of scalar field → vector field.

2) Distributed law
$\nabla\cdot(A+B) = \nabla\cdot A + \nabla\cdot B$

3) Any vector whose divergence is zero is called solenoidal vector. $\nabla\cdot\bar{A} = 0$.

? Determine the divergence of these vector fields.

a) $P = x^2yz\,\hat{a_x} + xz\,\hat{a_z}$

b) $Q = \rho\sin\phi\,\hat{a_\rho} + \rho^2 z\,\hat{a_\phi} + z\cos\phi\,\hat{a_z}$

c) $T = \frac{1}{r^2}\cos\theta\,\hat{a_r} + r\sin\theta\cos\phi\,\hat{a_\theta} + \cos\theta\,\hat{a_\phi}$

a) $P = x^2yz\,\hat{a_x} + xz\,\hat{a_z}$

$\nabla\cdot\bar{P} = \frac{\partial}{\partial x}(x^2yz) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(xz)$

$= 2xyz + x$

b) $Q = \rho \sin\phi \, \hat{a}_\rho + \rho^2 z \, \hat{a}_\phi + z\cos\phi \, \hat{a}_z$

$$\nabla \cdot Q = \frac{1}{\rho}\frac{\partial}{\partial \rho}\rho(\rho \sin\phi) + \frac{1}{\rho}\frac{\partial}{\partial \phi}\rho^2 z + \frac{\partial}{\partial z} z\cos\phi$$

$$= \frac{1}{R} \cdot 2R\sin\phi + \frac{1}{\rho} \times 0 + \cos\phi$$

$$= 2\sin\phi + \cos\phi$$

c) $T = \frac{1}{r^2}\cos\theta \, \hat{a}_r + r\sin\theta\cos\phi \, \hat{a}_\theta + \cos\theta \, \hat{a}_\phi$

$$\nabla \cdot \bar{T} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \cdot \frac{1}{r^2}\cos\theta\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(r\sin^2\theta\cos\phi) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\cos\theta$$

$$= \frac{1}{r^2} \cdot 0 + \frac{1}{r\sin\theta} r \cdot 2\sin\theta\cos\theta\cos\phi + \frac{1}{r\sin\theta} \cdot 0$$

$$= \underline{\underline{2\cos\theta \cdot \cos\phi}}$$

? Find the constant so that $\bar{V} = (x+3y)\hat{a}_x + (y-3)\hat{a}_y + (x+az)\hat{a}_z$ is solenoidal

$\bar{V}$ is solenoidal means $\nabla \cdot \bar{V} = 0$.

$$\frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-3) + \frac{\partial}{\partial z}(x+az) = 0$$

$$1 + 1 + a = 0$$

$$a = -2$$

? Find the divergence of $\bar{A}$ at pt $P[5, \pi/2, 1]$

cylindrical, where $\bar{A} = r z \sin\phi \, \hat{a}_r + 3 r z^2 \cos\phi \, \hat{a}_\phi$

$$\nabla \cdot \bar{A} = \frac{1}{r}\frac{\partial}{\partial r} r A_r + \frac{1}{r}\frac{\partial}{\partial \phi} A_\phi + \frac{\partial}{\partial z} A_z$$

$$= \frac{1}{r}\frac{\partial}{\partial r} r^2 z\sin\phi + \frac{1}{r}\frac{\partial}{\partial \phi}(3rz^2\cos\phi) + 0$$

$$= \frac{1}{r} \cdot 2rz\sin\phi + \frac{1}{r} \cdot 3rz^2 \cdot -\sin\phi$$

$$= 2z\sin\phi - 3z^2\sin\phi$$

At pt $P(5, \pi/2, +1)$

$$\nabla \cdot A = 2 \times 1 \times \sin(\pi/2) - 3 \times 1 \times \sin\pi/2$$

$$= 2 - 3 = \underline{\underline{-1}}$$

Curl

→ curl provides the measure of amt of rotation or angular momentum of a vector around the pt.

→ The curl of a vector field is the measure of the maximum circulation of the vector field around the boundary of an infinitesimal area per unit area as area tends to zero.

$$\boxed{\text{curl } \bar{A} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \oint_C A \, dl \, \hat{a}_n}$$

where ΔS is the area bounded by curve c and $\hat{a_n}$ is the unit vector normal to the surface. It indicates the direction along which maximum circulation occurs.

In vector analysis

$$\text{curl } \bar{A} = \nabla \times \bar{A}$$

$$\nabla \times \bar{A} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \text{ ; cartesian s/m}$$

$$\nabla \times \bar{A} = \frac{1}{\rho} \begin{vmatrix} \hat{a_\rho} & \rho \hat{a_\phi} & \hat{a_z} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix} \text{ , cylindrical}$$

$$\nabla \times \bar{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{a_r} & r\hat{a_\theta} & r\sin\theta \hat{a_\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

for spherical coordi.

note :-

If $\boxed{\nabla \times A = 0}$, then A is called irrotational.

also A is called conservative.

? Find the curl of the following vector fields

a) $P = x^2yz\,\hat{a_x} + xz\,\hat{a_z}$

b) $H = 2y\hat{a_x} + (z^2 - x^2)\hat{a_y} + 3y\,\hat{a_z}$

c) $B = 2\rho\cos\phi\,\hat{a_\rho} - 4\rho\sin\phi\,\hat{a_\phi} + 3\hat{a_z}$

a) $\text{curl } P = \nabla \times \bar{P} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & 0 & xz \end{vmatrix}$

$$= \hat{a_x}\left[\frac{\partial}{\partial y}(xz) - 0\right] - \hat{a_y}\left[\frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial z}(x^2yz)\right] + \hat{a_z}\left[0 - \frac{\partial}{\partial y}(x^2yz)\right]$$

$$= -\left[z - x^2y\right]\hat{a_y} - x^2z\,\hat{a_z}$$

$$= (x^2y - z)\hat{a_y} - x^2z\,\hat{a_z}$$

b) $\text{curl } H = \nabla \times \bar{H} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & z^2 - x^2 & 3y \end{vmatrix}$

$$= \hat{a_x}\left(\frac{\partial}{\partial y}3y - \frac{\partial}{\partial z}(z^2 - x^2)\right] - \hat{a_y}\left[\frac{\partial}{\partial x}3y - \frac{\partial}{\partial z}2y\right] + \hat{a_z}\left[\frac{\partial}{\partial x}(z^2 - x^2) - \frac{\partial}{\partial y}2y\right]$$

$$= \hat{a_x}[3-2z] - 0 + \hat{a_z}[-2x-2]$$

$$= [3-2z]\hat{a_x} - [2x+2]\hat{a_z}$$

c) curl $\bar{B} = \nabla \times \vec{B} = \frac{1}{\rho}\begin{vmatrix} \hat{a_\rho} & \rho\hat{a_\phi} & \hat{a_z} \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$

$$= \frac{1}{\rho}\begin{vmatrix} \hat{a_\rho} & \rho\hat{a_\phi} & \hat{a_z} \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ 2\rho\cos\phi & -4\rho^2\sin\phi & 3 \end{vmatrix}$$

$$= \frac{1}{\rho}\Big[\hat{a_\rho}\left(\frac{\partial}{\partial\phi}3 - \frac{\partial}{\partial z}4\rho^2\sin\phi\right] - \rho\hat{a_\phi}\left[\frac{\partial}{\partial\rho}3 - \frac{\partial}{\partial z}(2\rho\cos\phi)\right]$$
$$+ \hat{a_z}\left[\frac{\partial}{\partial\rho}(-4\rho^2\sin\phi) - \frac{\partial}{\partial\phi}(2\rho\cos\phi)\right]\Big]$$

$$= \frac{1}{\rho}\left[\hat{a_\rho}(0) - \rho\hat{a_\phi}[0] + \hat{a_z}[-8\rho\sin\phi + 2\rho\sin\phi]\right]$$

$$= (2\sin\phi - 8\sin\phi)\hat{a_z} = -6\sin\phi\,\hat{a_z}$$

? A vector $\bar{V}$ is irrotational, find the constants a, b, c. $\bar{V} = (x+2y+az]\hat{a_x} + (bx-3y-z)\hat{a_y} + (4x+cy+2z)\hat{a_z}$

$$\nabla \times V = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y+az & bx-3y-z & 4x+cy+2z \end{vmatrix}$$

$$= \hat{a_x}\left[\frac{\partial}{\partial y}(4x+cy+2z) - \frac{\partial}{\partial z}(bx-3y-z)\right] - \hat{a_y}\left[\frac{\partial}{\partial x}(4x+cy+2z) - \frac{\partial}{\partial z}(x+2y+az)\right] + \hat{a_z}\left[\frac{\partial}{\partial x}(bx-3y-z) - \frac{\partial}{\partial y}(x+2y+az)\right]$$

$$= \hat{a_x}[c+1] - \hat{a_y}[4-a] + \hat{a_z}[b-2]$$

$\nabla \times V = 0$

$c+1 = 0 \Rightarrow c = -1$

$4-a = 0 \Rightarrow 4 = a$

$b-2 = 0 \Rightarrow b = 2$

H.W ? vector field is given by $\bar{A} = yz\hat{a_x} + xz\hat{a_y} + xy\hat{a_z}$ S.T it is irrotation.

$$\nabla \times \bar{A} = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix}$$

$$= \hat{a_x}\left(\frac{\partial}{\partial y}(xy) - \frac{\partial}{\partial z}(xz)\right] - \hat{a_y}\left(\frac{\partial}{\partial x}xy - \frac{\partial}{\partial z}(yz)\right) + \ldots (xz)] - \frac{\partial}{\partial}(yz)]$$

$= \hat{a_x}[x-x] - \hat{a_y}[y-y] + \hat{a_z}(z-z)$

$= 0$

Divergence theorem / Gauss's Divergence theorem

D.v theorem states that total outward flux of a vector field A through the closed surface S is the same as the vol. integral of divergence of that vector.

$$\int_V \nabla \cdot \bar{A}\, dv = \oint_S A \cdot ds.$$

Proof (cartesian sm)

LHS

Divergence of any vector

$$\nabla \cdot \bar{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

volume integral on both sides

$$\iiint_V \nabla \cdot \bar{A}\, dv = \iiint_V \left[\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right] dv$$

$$= \iiint_V \left[\underset{A}{\frac{\partial A_x}{\partial x}} + \underset{B}{\frac{\partial A_y}{\partial y}} + \underset{C}{\frac{\partial A_z}{\partial z}}\right] dx\,dy\,dz \quad -(1)$$

$$A = \iiint_V \frac{\partial A_x}{\partial x} dx\,dy\,dz = \iint \left[\int \frac{\partial A_x}{\partial x} dx\right] dy\,dz$$

Let $\int_{x_1}^{x_2} \frac{\partial A_x}{\partial x} dx = A_{x_2} - A_{x_1} = A_x$

So $\iiint_V \frac{\partial A_x}{\partial x} dx\,dy\,dz = \iint_S A_x\, dy\,dz = \iint_S A_x\, ds_x.$

where $ds_x = dy\,dz$ is the x component of surface area ds.

Smly

$$B = \iiint_V \frac{\partial A_y}{\partial y} dx\,dy\,dz = \iint_S A_y\, dx\,dz$$

$$= \iint_S A_y\, ds_y$$

$$C = \iiint_V \frac{\partial A_z}{\partial z} dx\,dy\,dz = \iint_S A_z\, ds_z$$

substi in (1)

$$\iiint_V \nabla \cdot \bar{A}\, dv = \iint_S A_x\, ds_x + \iint_S A_y\, ds_y + \iint_S A_z\, ds_z$$

$$= \oint_S \bar{A} \cdot d\bar{s}$$

$$\iiint_V \nabla \cdot \bar{A}\, dv = \oint_S \bar{A} \cdot d\bar{s}.$$

* Divergence theorem applies to time varying as well as static fields. This theorem is used to convert volume integral of a divergence of a vector into a closed surface integral.

Using divergence theorem evaluate $\iint_S \bar{F}\cdot n\, ds$ where $\bar{F} = 2xy\,\hat{a_x} + y^2\,\hat{a_y} + 4yz\,\hat{a_z}$ and S is the surface of the cube bounded by $x=0, x=1, y=0, y=1$ & $z=0, z=1$.

By divergence theorem.

$$\iint_S \bar{F}\cdot n\,ds = \iiint_V \nabla\cdot\bar{F}\,dV$$

$$\nabla\cdot\bar{F} = \frac{\partial}{\partial x}F_x + \frac{\partial}{\partial y}F_y + \frac{\partial}{\partial z}F_z$$

$$= \frac{\partial}{\partial x}2xy + \frac{\partial}{\partial y}y^2 + \frac{\partial}{\partial z}4yz$$

$$= 2y + 2y + 4y = 8y$$

$$\iiint_V \nabla\cdot F\,dV = \iiint 8y\,dx\,dy\,dz$$

$$= \int_0^1\int_0^1\int_0^1 8y\,dx\,dy\,dz$$

$$= \int_0^1\int_0^1 8y\,x\Big]_0^1 dy\,dz$$

$$= \int_0^1\int_0^1 8y\cdot 1\cdot dy\,dz$$

$$= \int_0^1 \overset{4}{\cancel{8}}\left[\frac{y^2}{\cancel{2}}\right]_0^1 dz$$

$$= \int_0^1 4\,dz = 4z\Big]_0^1 = \underline{\underline{4}}$$

$$\iint_S \bar{F}\cdot n\,ds = \underline{4}$$

? Check validity of divergence theorem considering the field $\bar{D} = 2xy\,\hat{a_x} + x^2\,\hat{a_y}$ C/m^2 and the rectangular parallelopiped formed by planes $x=0, x=1$; $y=0, y=2$ & $z=0, z=3$.

$$\iint_S \bar{D}\cdot n\,ds = \iiint_V \nabla\cdot\bar{D}\,dV$$

RHS:

$$\nabla\cdot\bar{D} = \frac{\partial}{\partial x}D_x + \frac{\partial}{\partial y}D_y + \frac{\partial}{\partial z}D_z$$

$$= \frac{\partial}{\partial x}2xy + \frac{\partial}{\partial y}x^2 + \frac{\partial}{\partial z}0$$

$$= 2y + 0 = \underline{\underline{2y}}$$

$$\iiint_V \nabla\cdot\bar{D}\,dv = \int_{x=0}^{1}\int_{y=0}^{2}\int_{z=0}^{3} 2y\,dx\,dy\,dz$$

$$= \int_{x=0}^{1}\int_{y=0}^{2} 2y\,[z]_0^3\,dx\,dy$$

$$= \int_{x=0}^{1}\int_{y=0}^{2} 6y\,dx\,dy = \int_{x=0}^{1} \overset{}{\cancel{6}}_{3}\cdot\frac{y^2}{2}\Big]_0^2 dx$$

$$= \int_{x=0}^{1} 3\cdot[4-0]\,dx = \int_{x=0}^{1} 12\,dx = 12\cdot x\Big]_0^1$$

$$= \underline{\underline{12}}$$

L.H.S:

$$\iint_S \bar{D}\cdot n\,ds = \iint D\cdot\hat{a_x}\,dy\,dz + \iint D\cdot(-\hat{a_x})\,dy\,dz +$$

$$\iint D\cdot\hat{a_y}\,dx\,dz + \iint D\cdot(-\hat{a_y})\,dx\,dz +$$

$$\iint D\cdot\hat{a_z}\,dx\,dy + \iint D\cdot(-\hat{a_z})\,dx\,dy$$

$$\iint D\cdot \hat{a_x}\, dy\, dz = \int_0^3\int_0^2 [2xy\hat{a_x} + x^2\hat{a_y}]\cdot\hat{a_x}\, dy\, dz$$

$$= \int_0^3\int_0^2 2xy\, dy\, dz = \int_0^3 \left[\cancel{2}x\frac{y^2}{\cancel{2}}\right]_0^2 dz$$

$$= \int_0^3 x[4-0]\, dz = 4x[z]_0^3$$

$$= 12x = 12 \; (x=1)$$

$$\iint D\cdot(-\hat{a_x})\, dy\, dz$$

$$= \int_0^3\int_0^2 (2xy\hat{a_x} + x^2\hat{a_y})\cdot(-\hat{a_x})\, dy\, dz \Big|_{x=1}$$

$$= \int_0^3\int_0^2 -2xy\, dy\, dz = \int_0^3 -2x\left(\frac{y^2}{2}\right)_0^2 dz$$

$$= \int_0^3 -\cancel{2}x\left(\frac{4}{\cancel{2}}\right) dz = -4x\, z]_0^3$$

$$= -12x = 0 \quad [x=0]$$

$$\iint D\cdot\hat{a_y}\, dx\, dz = \int_0^3\int_0^1 (2xy\hat{a_x} + x^2\hat{a_y}]\cdot\hat{a_y}\, dx\, dz$$

$$= \int_0^3\int_0^1 x^2\, dx\, dz = \int_0^3 \frac{x^3}{3}]_0^1 dz$$

$$= \int_0^3 \frac{1}{3} dz = \frac{1}{3} z]_0^3 = \underline{\underline{1}}$$

$$\iint D\cdot -\hat{a_y}\, dx\, dz = \int_0^3\int_0^1 (2xy\hat{a_x} + x^2\hat{a_y}]\cdot -\hat{a_y}\, dx\, dz$$

Z plane constant since $\bar{D}$ lies in Z plane only x & y is vary

$$= \int_0^3\int_0^1 -x^2\, dx\, dz = \int_0^3 -\frac{x^3}{3}]_0^1 dz$$

$$= \int_0^3 -\frac{1}{3} dz = -\frac{1}{3} z]_0^3 = \underline{\underline{-1}}$$

$$\int_0^1\int_0^2 D\cdot\hat{a_z}\, dx\, dy = \int_0^1\int_0^2 (2xy\hat{a_x} + x^2\hat{a_y})\cdot\hat{a_z}\, dx\, dy$$

$$= 0$$

$$\iint D\cdot -\hat{a_z}\, dx\, dy = 0$$

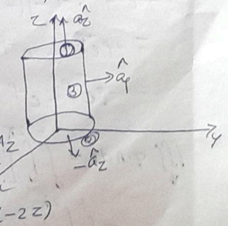
$$LHS = \iint_S \bar{D}\cdot n\, ds = 12 + 0 + 1 + -1 + 0 + 0$$

$$= 12$$

LHS = RHS. - Divergence theorem is verified.

? Given $\bar{A} = 30e^{-\rho}\hat{a_\rho} - 2z\hat{a_z}$ in cylinderical coordinates. Evaluate both sides of divergence theorem for volume enclosed by $\rho = 2$, $z = 0$ & $z = 5$.

D.V Theorem states that



$$\iiint_V \nabla\cdot A\, dv = \oiint_S A\cdot ds$$

LHS

$$\nabla\cdot A = \frac{1}{\rho}\frac{\partial}{\partial\rho}\rho A_\rho + \frac{1}{\rho}\frac{\partial}{\partial\phi}A_\phi + \frac{\partial}{\partial z}A_z$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}\rho\cdot 30e^{-\rho} + 0 + \frac{\partial}{\partial z}(-2z)$$

$$= \frac{1}{\rho}[\rho\cdot 30e^{-\rho} \times -1 + 30e^{-\rho}\cdot 1] - 2$$

$$= \frac{-\rho 30 e^{-\rho} + 30 e^{-\rho} - 2\rho}{\rho}$$

$$= \frac{30 e^{-\rho} - 30 \rho e^{-\rho} - 2\rho}{\rho}$$

$$\iiint_V \nabla \cdot A \, dv = \iiint \frac{30 e^{-\rho} - 30 \rho e^{-\rho} - 2\rho}{\rho} \cdot \rho \, d\rho \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} \int_0^2 (30 e^{-\rho} - 30 \rho e^{-\rho} - 2\rho) \, d\rho \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} -30 e^{-\rho} - 30 \left[\rho e^{-\rho} - \int e^{-\rho} \times 1 \, d\rho \right] - 2 \frac{\rho^2}{2} \Big] \, d\phi \, dz$$

$\int u v \, dx = u \int v \, dx - \int (u' \int v \, dx) \, dx$

$$\int \rho e^{-\rho} d\rho = \rho \int e^{-\rho} d\rho - \int \left(1 \int e^{-\rho} d\rho\right) d\rho$$

$$= \rho e^{-\rho} \cdot -1 - \int (-e^{-\rho}) \, d\rho$$

$$= -\rho e^{-\rho} + e^{-\rho} \times -1$$

$$= -\rho e^{-\rho} - e^{-\rho}$$

$$\iiint_V \nabla \cdot A \, dv = \int_0^5 \int_0^{2\pi} \left[-30 e^{-\rho} - 30 \left[-\rho e^{-\rho} - e^{-\rho} \right] - 2 \frac{\rho^2}{2} \right]_0^2$$

$$= \int_0^5 \int_0^{2\pi} \cancel{-30 e^{-\rho}} + 30 \rho e^{-\rho} + \cancel{30 e^{-\rho}} - \rho^2 \Big]_0^2 \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} 30 \rho e^{-\rho} - \rho^2 \Big]_0^2 \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} (30 \times 2 e^{-2} - 4) \, d\phi \, dz = \int_0^5 \int_0^{2\pi} 4.12 \, d\phi \, dz$$

$$= \int_0^5 4.12 \, \phi \Big]_0^{2\pi} dz = \int_0^5 4.12 \times 2\pi \, dz$$

$$= 4.12 \times 2\pi \, z \Big]_0^5 =$$

$$= \underline{\underline{129.43}}$$

RHS

$$\oint\!\!\oint A \cdot ds = \iint_{①} A \cdot ds + \iint_{②} A \cdot ds + \iint_{③} A \cdot ds.$$

$$\iint_{①} A \cdot ds = \iint \left[30 e^{-\rho} \hat{a}_\phi - 2z \hat{a}_z \right] \cdot \rho \, d\rho \, d\phi \, \hat{a}_z$$

$$= \int_0^{2\pi} \int_0^2 -2z \rho \, d\rho \, d\phi = \int_0^{2\pi} -2z \frac{\rho^2}{2} \Big]_0^2 d\phi$$

$$= \int_0^{2\pi} -\cancel{2} z \cdot \frac{4}{\cancel{2}} \, d\phi = -4z \, \phi \Big]_0^{2\pi} = -8\pi z \, [z=5$$

$$= -\underline{\underline{40\pi}}$$

$$\iint_{②} A \cdot ds = \iint (30 e^{-\rho} \hat{a}_\phi - 2z \hat{a}_z) \cdot \rho \, d\rho \, d\phi \, -\hat{a}_z$$

$$= \int_0^{2\pi} \int_0^2 +2z \rho \, d\rho \, d\phi = \int_0^{2\pi} \cancel{2} z \frac{\rho^2}{\cancel{2}} \Big]_0^2 d\phi$$

$$= \int_0^{2\pi} z \cdot 4 \, d\phi = 4z \cdot 2\pi$$

$$= 8\pi z$$

$$= 0 \; ; \; z = 0$$

$$\iint_{③} A\cdot ds = \iint \left[30e^{-\rho}\hat{a}_\phi - 2z\hat{a}_z\right]\cdot \rho\, d\phi\, dz\, \hat{a}_\rho$$

$$= \int_0^5\int_0^{2\pi} 30e^{-\rho}\cdot \rho\, d\phi\, dz$$

$$= \int_0^5 30e^{-\rho}\cdot \rho\cdot \phi\Big]_0^{2\pi} dz$$

$$= \int_0^5 30e^{-\rho}\cdot \rho\cdot 2\pi\, dz$$

$$= 60\pi e^{-\rho}\cdot \rho\cdot z\Big]_0^5 = 300\pi \underset{2}{\rho} e^{-\rho}$$

$$= 600\pi e^{-2};\ \rho=2$$

$$= 81\cdot 2\pi$$

$$\oiint A\cdot ds = -40\pi + 0 + 81\cdot 2\pi$$

$$= 129\cdot 43$$

LHS = RHS

1. Verify divergence theorem for the vector $A = 4x\,\hat{a}_x - 2y^2\,\hat{a}_y + z^2\,\hat{a}_z$ taken over the cube bounded by $x=0, x=1, y=0, y=1$, and $z=0, z=1$

Divergence theorem is given by,

$$\iiint_V \nabla\cdot A\, dv = \oint\!\!\oint_S A\cdot ds$$

LHS :

$$\nabla\cdot A = \frac{\partial}{\partial x} A_x + \frac{\partial}{\partial y} A_y + \frac{\partial}{\partial z} A_z$$

$$= \frac{\partial}{\partial x} 4x + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z} z^2$$

$$= 4 - 4y + 2z$$

$$\iiint_V \nabla\cdot A\, dv = \int_0^1\int_0^1\int_0^1 4 - 4y + 2z \; dx\,dy\,dz$$

$$= \int_0^1\int_0^1 \left[4x - 4xy + 2zx\right]_0^1 dy\,dz$$

$$= \int_0^1\int_0^1 4 - 4y + 2z \; dy\,dz$$

$$= \int_0^1 \left[4y - \frac{4y^2}{2} + 2zy\right]_0^1 dz$$

$$= \int_0^1 \left[4 - 2 + 2z\right] dz$$

$$= \int_0^1 2 + 2z \; dz$$

$$= 2z + \frac{2z^2}{2} \Big]_0^1$$

$$= 2 + 1 = \underline{3}$$

$$\iint A \cdot ds = \iint A \cdot \hat{a}_x \, dy\,dz + \iint A \cdot (-\hat{a}_x) \, dy\,dz + \iint A \cdot \hat{a}_y \, dx\,dz$$

$$+ \iint A \cdot (-\hat{a}_y) \, dx\,dz + \iint A \cdot \hat{a}_z \, dx\,dy + \iint A \cdot (-\hat{a}_z) \, dx\,dy$$

$$\iint A \cdot \hat{a}_x \, dy\,dz = \int_0^1\int_0^1 [4x\,\hat{a}_x - 2y^2\hat{a}_y + z^2\hat{a}_z] \cdot \hat{a}_x \, dy\,dz$$

$$= \int_0^1\int_0^1 4x \, dy\,dz$$

$$= \int_0^1 4x\,y\Big]_0^1 dz$$

$$= \int_0^1 4x \, dz$$

$$= 4xz\Big]_0^1$$

$$= 4x.$$

$$= 4 \quad [x = 1]$$

$$\iint A \cdot -\hat{a}_x \, dy\,dz = \int_0^1\int_0^1 [4x\,\hat{a}_x - 2y^2\hat{a}_y + z^2\hat{a}_z] \cdot -\hat{a}_x \, dy\,dz$$

$$= \int_0^1\int_0^1 -4x \, dy\,dz$$

$$= \int_0^1 -4x\,y\Big]_0^1 dz$$

$$= \int_0^1 -4x \, dz$$

$$= -4xz \Big]_0^1$$

$$= -4x$$

$$= 0 \quad [x=0]$$

$$\iint A \cdot \hat{a}_y \, dx\, dz = \int_0^1\int_0^1 [4x\,\hat{a}_x - 2y^2\hat{a}_y + z^2\hat{a}_z] \cdot \hat{a}_y \, dx\, dz$$

$$= \int_0^1\int_0^1 -2y^2 \, dx\, dz$$

$$= \int_0^1 \Big[-2y^2 x\Big]_0^1 dz$$

$$= \int_0^1 -2y^2 \, dz$$

$$= -2y^2 z\Big]_0^1$$

$$= -2y^2$$

$$= -2 \quad [y=1]$$

$$\iint A \cdot \bar{a}_y \, dx\, dz = \int_0^1\int_0^1 [4x\,\hat{a}_x - 2y^2\hat{a}_y + z^2\hat{a}_z] \cdot \hat{a}_y \, dx\, dz$$

$$= \int_0^1\int_0^1 2y^2 \, dx\, dz$$

$$= 0 \quad ; [y=0.]$$

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$$\iint A \cdot \hat{a}_z \, dx\,dy = \int_0^1\int_0^1 [4x\,\hat{a}_x - 2y^2\hat{a}_y + z^2\hat{a}_z] \cdot \hat{a}_z \, dx\,dy$$

$$= \int_0^1\int_0^1 z^2 \, dx\,dy$$

$$= \int_0^1\int_0^1 z^2[x]_0^1 \, dy$$

$$= \int_0^1 z^2 \, dy$$

$$= z^2[y]_0^1$$

$$= z^2 = 1 \quad ; \{z = 1\}$$

$$\iint A \cdot -\hat{a}_z \, dx\,dy = \int_0^1\int_0^1 [4x\,\hat{a}_x - 2y^2\hat{a}_y + z^2\hat{a}_z] \cdot -\hat{a}_z \, dx\,dy$$

$$= \int_0^1\int_0^1 -z^2 \, dx\,dy$$

$$= -z^2 \int_0^1 [x]_0^1 \, dy$$

$$= -z^2 \, [y]_0^1 = -z^2 = 0 \quad \{z = 0\}$$

$$RHS = \oiint A \cdot ds = 4 + 0 + -2 + 0 + 1 + 0$$

$$= 3$$

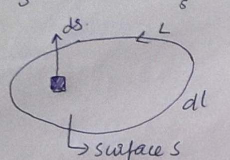
LHS = RHS

Hence divergence theorem is verified

Stokes Theorem

Stokes theorem relate the line integral to a surface integral. It states that the circulation of a vector field $\bar{A}$ around a closed path L is equal to the surface integral of the curl of $\bar{A}$ over the open surface S bounded by L.

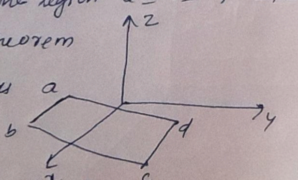
$$\oint_L A \cdot dl = \iint_S (\nabla \times A) \cdot ds = \iint_S (\nabla \times \bar{A})\, \hat{a_n}$$



? Evaluate both sides of stokes theorem for the field $\bar{H} = 6xy\,\hat{a_x} - 3y^2\,\hat{a_y}$ A/m & the rectangular path around the region $2 \le x \le 5, -1 \le y \le 1, z = 0$

Acc. to stokes theorem

$$\oint_L H \cdot dl = \iint_S (\nabla \times H) \cdot ds$$



LHS

$$\oint_L H \cdot dl = \int_a^b + \int_b^c + \int_c^d + \int_d^a H \cdot dl$$

~~Lhs~~

$$\int_a^b H \cdot dl = \int_{a=2}^{5} (6xy\,\hat{a_x} - 3y^2\,\hat{a_y}] \cdot dx\,\hat{a_x}$$

$$= \int_2^5 6xy\,dx = 6y\frac{x^2}{2}\Big]_2^5$$

$$= 3y[25-4] = 63y$$

for line ab value of $y = -1$

$$\int_a^b H \cdot dl = 63 \times -1 = \underline{-63}$$

$$\int_b^c H \cdot dl = \int_{-1}^{1} (6xy\,\hat{a_x} - 3y^2\,\hat{a_y}) \cdot dy\,\hat{a_y}$$

$$= \int_{-1}^{1} -3y^2\,dy = -3\frac{y^3}{3}\Big]_{-1}^{1}$$

$$= -1[1 - -1] = \underline{\underline{-2}}$$

$$\int_c^d \bar{H} \cdot dl = \int_5^2 (6xy\,\hat{a_x} - 3y^2\,\hat{a_y}) \cdot dx\,\hat{a_x}$$

$$= \int_5^2 6xy\,dx = 6y\frac{x^2}{2}\Big]_5^2 = 3y[4-25]$$

$$= -63y = -63 \times 1 (y=1)$$

$$= -63$$

$$\int_d^a H \cdot dl = \int_1^{-1} (6xy\,\hat{a_x} - 3y^2\,\hat{a_y}] \cdot dy\,\hat{a_y}$$

$$= -3\frac{y^3}{3}\Big]_1^{-1} = -[-1-1] = 2$$

$$\int H \cdot dl = -63 - 2 - 63 + 2 = \underline{\underline{-126}}$$

RHS

$$\nabla \times H = \begin{vmatrix} \hat{a_x} & \hat{a_y} & \hat{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy & -3y^2 & 0 \end{vmatrix} = \hat{a_x}\left[0 - \frac{\partial}{\partial z}(-3y^2)\right] - \hat{a_y}\left[0 - \frac{\partial}{\partial z}(6xy)\right] + \hat{a_z}\left[\frac{\partial}{\partial x}(-3y^2) - \frac{\partial}{\partial y}(6xy)\right]$$

$$= \hat{a_x}(0) - \hat{a_y}(0) + \hat{a_z}(-6x)$$

$$= -6x\,\hat{a_z}$$

$$\iint_S \nabla \times H \cdot ds = \iint -6x\,\hat{a_z} \cdot dx\,dy\,\hat{a_z}$$

$$= \int_{x=2}^{5}\int_{y=-1}^{1} -6x\,dx\,dy$$

$$= \int_{-1}^{1} \left.-\cancel{6}^{3}\frac{x^2}{\cancel{2}}\right]_2^5 dy = \int_{-1}^{1} -3\left[25-4\right] dy$$

$$= -63\,y\Big]_{-1}^{1} = \underline{\underline{-126}}$$

LHS = RHS